

HEMA HARSHA VARDHAN PEELA

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SUMMARY

Data Engineer with **5+ years** of experience designing, implementing, and optimizing **large-scale data solutions** across **financial, healthcare, and enterprise domains**. Expert in **end-to-end ETL/ELT pipelines**, **cloud data platforms (AWS, Azure, GCP)**, **distributed processing (Spark, Python)**, and **data lakes**, managing hundreds of millions of records efficiently. Skilled in **generative AI analytics**, **knowledge graphs**, **real-time dashboards**, and **machine learning integration** to deliver actionable **business insights**. Strong in **data governance**, **security**, and **compliance**, with a track record of **mentoring teams**, **driving automation**, and enabling **data-driven decision-making** for strategic growth.

PROFESSIONAL EXPERIENCE

Data Engineer | Wells Fargo | San Francisco, CA, USA **June 2025 – Present**

- Architected and deployed scalable data pipelines using AWS Glue, S3, Athena, and Spark, processing over 500 million transactional records monthly to support enterprise-wide analytics initiatives.
- Engineered and automated complex ETL workflows leveraging PySpark and Python, reducing manual data processing effort by 1,200+ hours annually while ensuring 99.9% data accuracy across downstream systems.
- Designed and maintained knowledge graphs and ontologies with Neo4j and RDF/OWL frameworks, improving cross-departmental data discovery and accelerating report generation by 3x.
- Implemented advanced data security measures including IAM policies, KMS encryption, and Lake Formation access controls, safeguarding 1TB+ sensitive financial data and ensuring compliance with regulatory standards.
- Streamlined data processing workflows across distributed systems and cloud platforms, increasing pipeline efficiency by 2x while supporting multiple high-volume projects simultaneously.
- Collaborated with cross-functional teams to troubleshoot data issues, optimize performance, and drive best practices in ETL, cloud deployment, and data governance.

Data Engineering Intern | YourBook Team | Lubbock, TX, USA **November 2024 – May 2025**

- Assisted in building and maintaining 10+ SQL-based KPI reports and executive dashboards, enabling leadership to monitor key business metrics effectively.
- Reviewed large datasets to identify and log 50+ data issues and inconsistencies, preparing findings for validation by senior analysts.
- Supported data validation and quality checks in reporting workflows to ensure accuracy before stakeholder review.
- Documented reporting logic and data definitions to support consistent reporting and team knowledge sharing.

Data Engineer | Cognizant | Bengaluru, India **January 2021 – June 2023**

- Orchestrated and optimized large-scale ETL/ELT pipelines managing 50+TB of critical enterprise data, delivering timely, actionable insights across 10+ business divisions.
- Spearheaded and optimized data pipelines using Talend, Informatica IICS, and DataStage, resolving 30+ critical job failures monthly and enhancing overall pipeline reliability.
- Administered and scaled Azure Data Services (Data Factory, Synapse, Databricks, Azure SQL Database) to support real-time analytics for 10+ business divisions.
- Automated and orchestrated Python, SQL, and Spark-based data workflows, processing millions of transactions daily while maintaining compliance and data governance standards.
- Mentored and coached 5+ junior engineers, improving team efficiency, knowledge transfer, and adoption of best practices in ETL, data modeling, and cloud deployment.
- Partnered cross-functionally with business analysts, architects, and SRE teams to troubleshoot high-priority data issues, achieving zero SLA violations across critical workflows.
- Documented and communicated technical designs, data flows, and integration procedures to stakeholders, driving knowledge sharing and onboarding of new team members within 2 weeks.

Junior Data Engineer | Cipla | India **November 2019 – December2020**

- Integrated and validated clinical and operational data from 3+ hospital systems using SQL and AWS Glue, supporting analysis of 150,000+ patient and service records for actionable operational insights.

- Developed and automated ETL pipelines to improve data ingestion and processing efficiency, enhancing refresh reliability and reducing manual effort by ~60 hours per month.
- Created Power BI dashboards to track service utilization, patient throughput, and operational KPIs, reviewed weekly by 20+ clinical and operations stakeholders for decision-making.
- Optimized multi-hospital data pipelines handling 150,000+ records monthly, ensuring data integrity and governance, while accelerating reporting and enabling cross-departmental analytics.

PROJECTS

Generative AI for Business Insights and Demand Forecasting | *Python, SQL, ETL, GCP (BigQuery), Generative AI (GPT), Power BI/Tableau*

- Built and optimized ETL pipelines to consolidate and transform 500,000+ business transactions using Python, SQL, and GCP BigQuery, ensuring data integrity and delivering actionable analytics.
- Leveraged Generative AI models to produce narratives, detect anomalies, and forecast demand in interactive dashboards, enabling actionable, data-driven decisions.

Cloud-Based Real-Time Fraud Detection & Analytics Platform | *SQL, Python, Power BI, Tableau, Data Analytics, Fraud Detection*

- Revamped analytics workflows to monitor 100,000+ daily transactions, flagging exceptions and risks in near real-time.
- Created Power BI and Tableau dashboards with fraud detection models and documented 20+ data flows for scalable reporting.

TECHNICAL SKILLS

Programming & Scripting: Python (PySpark, Pandas, NumPy), SQL (T-SQL, PostgreSQL, BigQuery SQL, Redshift, Snowflake), Shell Scripting, R, Excel VBA, Python & SQL Automation

Data Engineering & ETL: ETL/ELT Pipeline Design, Optimization & Automation (AWS Glue, Talend, Informatica IICS, DataStage, SSIS, Azure DataFlow, Google DataFlow), Big Data Processing (Spark, AWS EMR, Azure Databricks), Data Integration, Profiling, Validation & Reconciliation, OLAP & Data Modeling (Star/Snowflake schema, Data Warehousing, Redshift, Snowflake, BigQuery), Data Lake Management (AWS S3, Azure Data Lake, Lake Formation), Workflow Automation & Pipeline Orchestration

Cloud Platforms & Services: AWS: S3, EC2, Lambda, IAM, Glue, Athena, Redshift, Auto Scaling | Azure: Azure Data Lake, Azure DevOps | GCP: BigQuery, DataFlow

Data Analytics & Visualization: Power BI, Tableau, KPI Dashboards, Real-time Reporting, Advanced Excel (Pivot Tables, Lookups, Macros), Visualization Libraries (Matplotlib, Seaborn, Plotly), Business Insights & Operational Analytics

Machine Learning & AI: Generative AI (GPT-based models), Predictive Analytics & Forecasting, Supervised & Unsupervised Learning (Regression, Classification, Clustering, Decision Trees, Random Forest, Neural Networks), Anomaly Detection, NLP & Text Analytics (Sentiment Analysis, NLTK), AI-assisted Data Validation

Data Governance & Quality: Data Standardization & Validation, Data Lineage & Dictionaries, Data Security & Access Controls, Compliance & Operational Analytics

DevOps & Collaboration Tools: Git, Jenkins, JIRA, Confluence, Zephyr Scale, Jupyter Notebook, Google Colab, Agile/Scrum, SDLC

EDUCATION

Texas Tech University | TX, USA

Masters in Computer Science

August 2023 – May 2025

GPA 3.7/4.0